

Sr no	Category	Description
1	Hardware	Automatic regulation of valves for release of water based upon soil moisture availability in the root zone of the crop, using artificial intelligence, in a piped and
		Monitoring through AI-based Remote Access Vehicle.
		Automated Public Lighting.
		Active Prosthetic ankle and adaptive equipment for bike riding in lower limb amputees.
		Drone-based surveillance system for vessels in port areas and encroachments.
		A system of IoT devices to prevent under-loading/overloading of railway wagons.
		Development of a Telematic control unit for capturing vital data of a vehicle without using company-fitted devices.
		Development of Smart Toilet.
		Developing a system for Patient Care in the Health Sector.
		Call for cost-effective ways of making water sources for piped drinking water supply sustainable in rural areas.
		Difficulty in operating heavy earth-moving machinery during the rainy season due to extremely poor visibility conditions.
		Mines operation, especially haulage of dumpers, done through operators even in their absence.
		Unpredictable wear and tear of cable belt conveyor rope and belt leading to frequent stoppage of a single-line mine production system.
		Innovative Solution for Reducing AT&C Losses due to Power Pilferage.
		Development of Small Scale Wind Energy Device.
		Despite prohibition, hazardous cleaning of sewers and septic tanks continues without safety kits, safety devices, and adherence to safety protocols.
		One-stop solution for monitoring dairy plant energy consumption, hygiene, and packaging waste collection from consumers.
		Geo-tagging of plantation in the catchment area of a hydro project.
		Robotics for inspection of abrasion/corrosion of underwater equipment/parts and further repair.
		Detection of embedded Malware/Trojan in hardware devices used in the Power Sector.
		Development of technology for manufacturing a mind-controlled BIONIC hand with a sense of touch.
		Uncontrolled growth of water hyacinth in lakes.
		Low-cost Myoelectric Prosthetic Arm.
		Development of Augmentative and Alternative Communication (AAC) in the Indian context.
		Drone-based Intelligent Magnetic Sensing System and Metallic Anomaly Detection System.

		Call for low-cost desalination technology for Lakshadweep and inland saline water.
		Technological solutions for early decomposition of fecal matter.
		Call for Toilet Technology.
		Lowest Cost Smart Board - A Seamless Teaching Solution.
		Robotics Device for Borewell Rescue Operation.
		Smart Glass Cleaning Robot, which can climb, clean, and carry liquid cleaners using Autodesk Fusion 360.
		Design of RF Up/Downconverter for signals using GNU Radio and SDRs.
		Student Innovation.
		Green options for milk packaging (low-cost, environment-friendly, and extended shelf-life packaging).
		Automatic Drug Dispenser.
		Frequent dislodgement of belt conveyors along hilly terrain for various reasons.
		Real-time knowledge of ore body being mined out.
		Centralized Monitoring System for Street Light Fault Detection and Location.
		Effective management of construction and demolition (C&D) waste.
		Design an IoT-based system to track and manage food waste in grocery stores and restaurants.
		Identification of victims buried under avalanches using artificial intelligence and the Internet of Things.
		Artificial intelligence-based valves for release of water based on soil moisture availability in the root zone of the crop, in a micro irrigation network.
		Dashboard for real-time monitoring of construction projects.
		Develop a system for the awareness of people who do not follow traffic rules at cross roads.
		Develop a system to detect vehicles using their number plate and inform the police.
		Design a novel technology for early detection and mitigation of avalanches.
		Artificial intelligence-based smart attendance system for students and employees.
		Smart system to catch motorcycle and car thieves using artificial intelligence and machine learning.
		Remote monitoring of health parameters for medical consultation and home-admission.
		Design and develop a solution leveraging social robots to provide continuous, round-the-clock care for patients.
		Automated warehouse sorting: Develop a robotic arm solution capable of efficiently sorting and organizing diverse items in a warehouse setting, optimizing inventory management processes.

		Assistive robotic arm for elderly care: Create a robotic arm system that assists elderly individuals with daily tasks such as fetching objects, opening containers, or providing support for mobility.
		Precision agriculture: Design a robotic arm equipped with sensors and actuators to perform precise agricultural tasks such as planting seeds, applying fertilizers, or harvesting crops in a controlled environment.
		Medical rehabilitation assistance: Develop a robotic arm solution that aids in the rehabilitation process for patients recovering from injuries or surgeries, assisting with exercises and physical therapy routines.
		Interactive educational tool: Create an educational robotic arm platform that engages students in learning about robotics, automation, and engineering concepts through interactive demonstrations and hands-on programming.
		Personalized manufacturing assistant: Design a robotic arm system capable of assisting small-scale manufacturers or hobbyists in the production process, offering customizable features for different manufacturing tasks.
		Autonomous home assistant: Develop a robotic arm integrated with AI capabilities to serve as a versatile home assistant, capable of performing tasks such as cleaning, organizing, or assisting with household chores.
		Remote inspection and maintenance: Create a robotic arm solution equipped with cameras and sensors for remote inspection and maintenance of infrastructure or industrial equipment in hazardous or hard-to-reach environments.
		Environmental cleanup and conservation: Design a robotic arm system capable of collecting and disposing of waste materials or pollutants in natural environments, contributing to environmental cleanup and conservation efforts.
		Artistic expression and performance: Develop a robotic arm platform that serves as a creative tool for artists, performers, or designers, enabling expressive movement and interactive installations in artistic or entertainment settings.
		Monitoring of remote access vehicles using an AI-based system.
		Development of a telematic control unit for capturing vital data of a vehicle without using a company-fitted telemetry data port.
		IoT-based safety systems for public places/transportation.
		IoT with emerging technologies (AI, ML, XR, Smart Sensors, etc.) for smart farming.
		IoT with emerging technologies for healthcare applications.
		IoT with emerging technologies for industrial automation.
		IoT with emerging technologies for electric vehicles.
		IoT-integrated wearable devices.
		IoT with emerging technologies for sustainable smart cities.

		IoT with emerging technologies in education.
		Smart Education, a Concept that Describes learning in digital age.it enables learner to learn more effectively, efficently, flexibly and comfortably.
		Centralized Monitoring System for Street Light Fault Detection and Location
		Design of RF Up/Downconverter for signals using GNU Radio and SDRs.
		Students to use Autodesk Fusion 360 to research and develop a design on Smart Glass Cleaning Robot, which can climb, clean, and carry liquid cleaners.
		Drone based Intelligent Magnetic sensing system and Metallic anomaly detection.
		Detection of embedded Malware/ Trojan in hardware devices used in Power Sector
		Development of Small Scale Wind energy device
		A system of IoT Devices to prevent under-loading / overloading of Railway
		Monitoring through AI Based Remote Access Vehicle
		Developing an AI-powered energy management system for industrial commercial facilities to optimize energy
		Monitoring through AI Based Remote Access Vehicle